

QWIM MARKET REGIME DETECTION AND REGIME-AWARE PORTFOLIO ALLOCATION



Medium–Long Term Wealth Management Application

Phase 2

FE800

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Why Regimes Matter for Medium–Long Term Wealth Allocation

- Financial markets are **not stationary**.
- Over time, we observe persistent structural shifts in:
 - Expected returns (*Nystrup et al., 2017*)
 - Volatility and tail risk (*Nystrup et al., 2017; Aydinhan et al., 2024*)
 - Cross-asset correlations (*Nystrup et al., 2017; Aydinhan et al., 2024*)
 - Credit conditions (*Aydinhan et al., 2024*)
 - Inflation (*Pomorski, 2024*)
 - Monetary and macro cycles (*Pomorski, 2024; Hirsä et al., 2024*)
- **Our thesis:** Portfolio allocation should adapt to persistent regime shifts, while controlling turnover and transaction costs.
 - Objective:
 - Improve drawdown resilience
 - Enhance smooth compounding
 - Preserve capital across cycles



Data & Feature Matrix

49 Daily Macro & Market Signals (2001–2025) · All features lagged ≥ 1 day before use

Equity Market (8)

SPX return, vol 20d, drawdown, momentum, MA signal

Credit (5)

HY spread, IG spread, HY/IG changes, HY-IG differential

Macro & Inflation (7)

CPI change, CPI YoY, inflation regime, ISM, UMich, CB confidence

Safe Haven (6)

Gold return, GLD/SPX ratio, ratio return, momentum 21d

Volatility & Sentiment (5)

VIX level, VIX change, VIX 3m–spot term structure

Rates & Curve (8)

GT2, GT10, GT30, slope 10–2, slope 30–10, yield change

Global Risk (6)

DXY level, return, momentum 63d, vol 21d, DXY–SPX corr

Monetary (4)

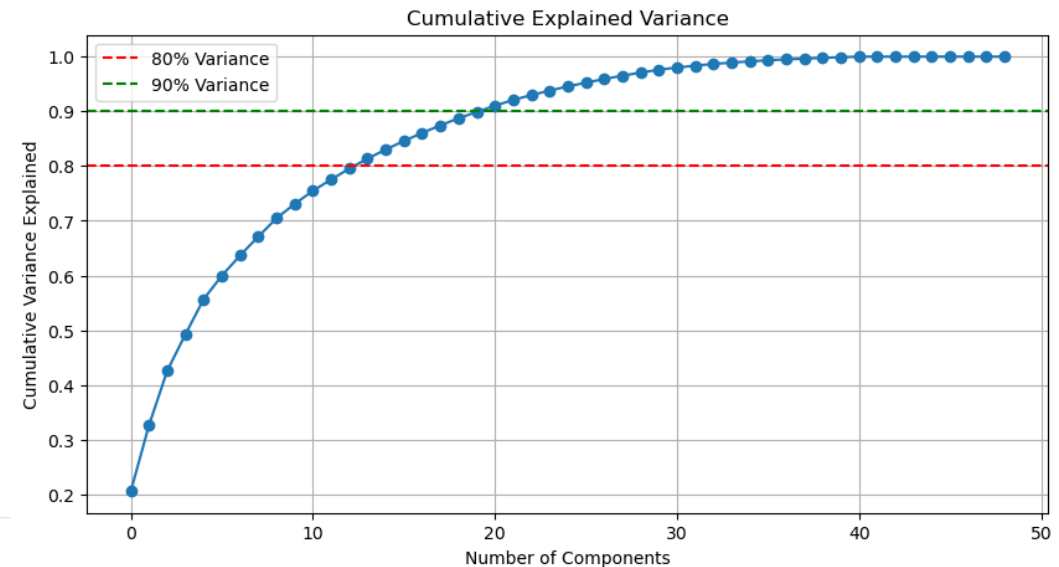
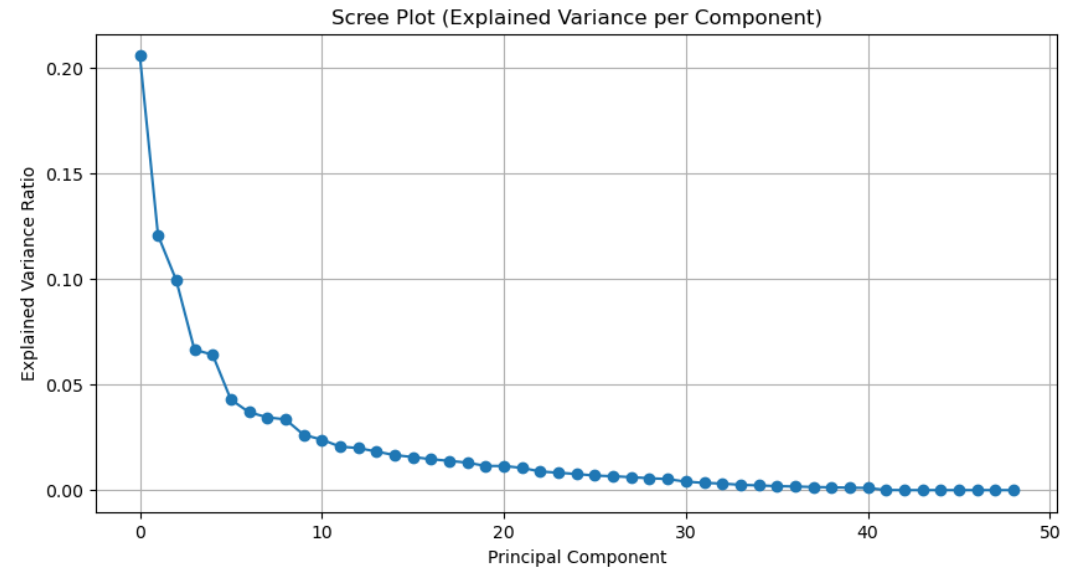
M2 growth 1m/3m/YoY lag, NBER recession indicator lag

+ 3 augmented features: `_yield_trend_63d` · `_spx_cum_21d` · `_vix_yield_interaction` (added before PCA to help GMM separate rate-hike stress from crash stress)

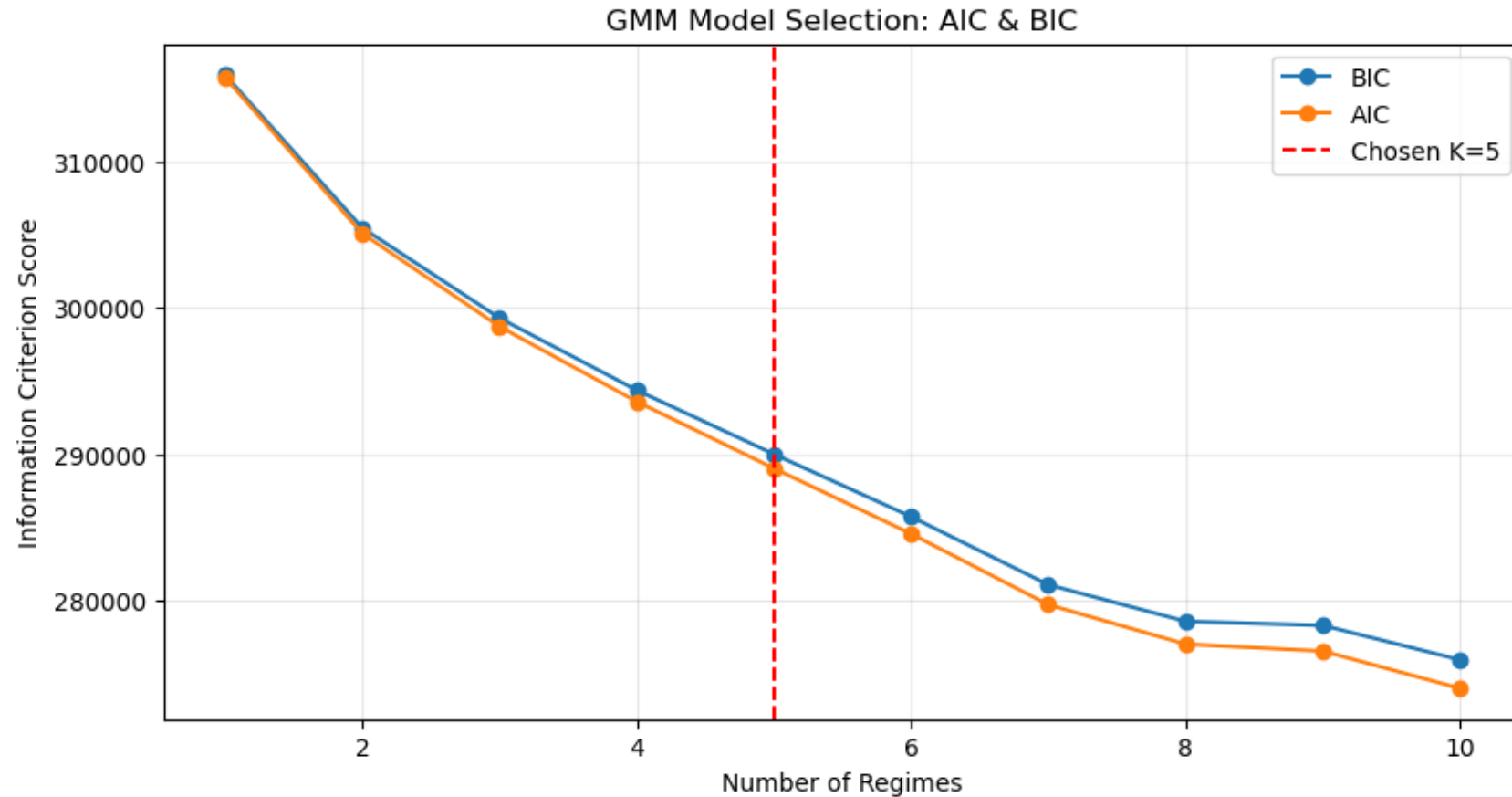


Principal Component Analysis (PCA)

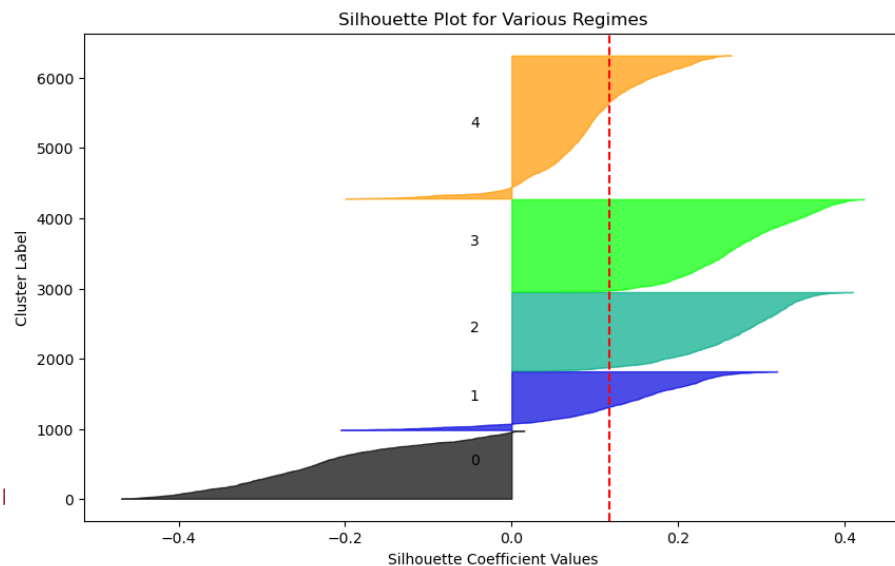
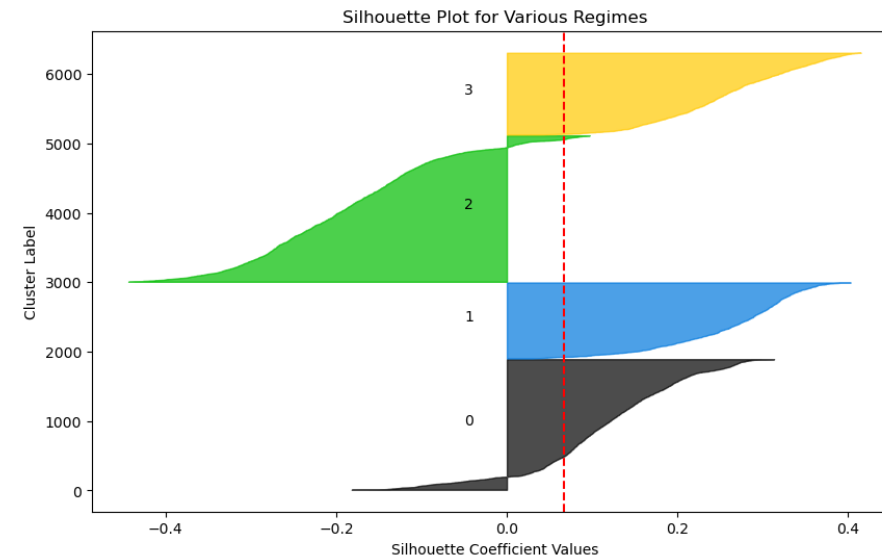
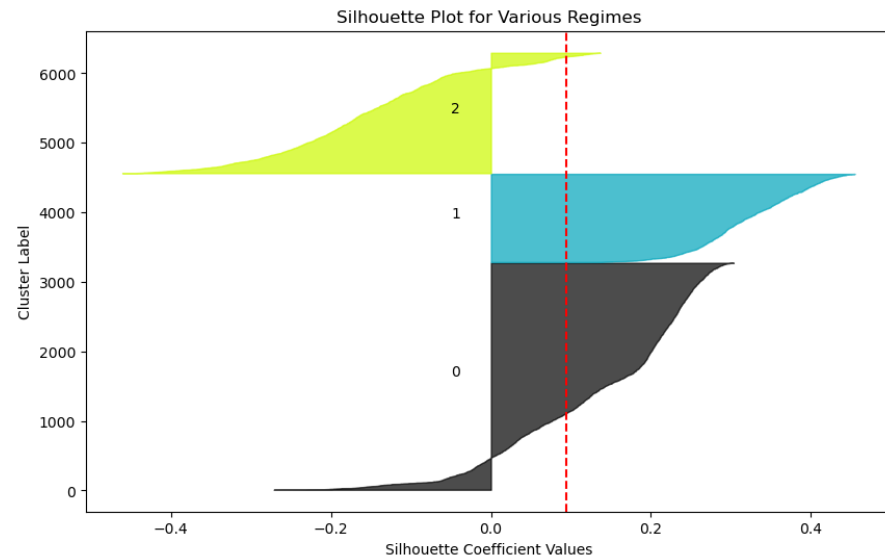
- Features are standardized before PCA
- PCA reduces dimensionality while preserving structure
- Components for $\geq 80\%$ variance: 14
- Components for $\geq 90\%$ variance: 21



GMM Model Selection: AIC & BIC

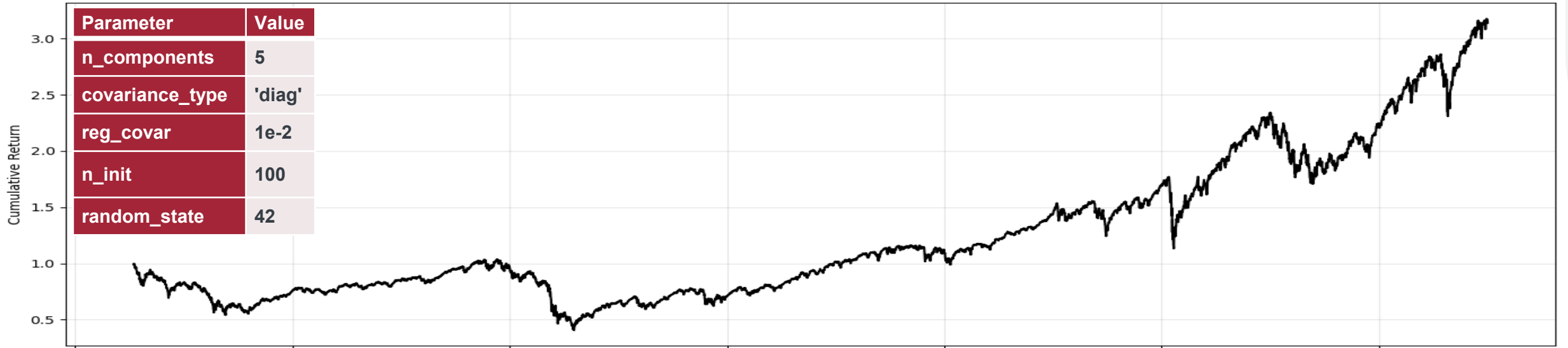
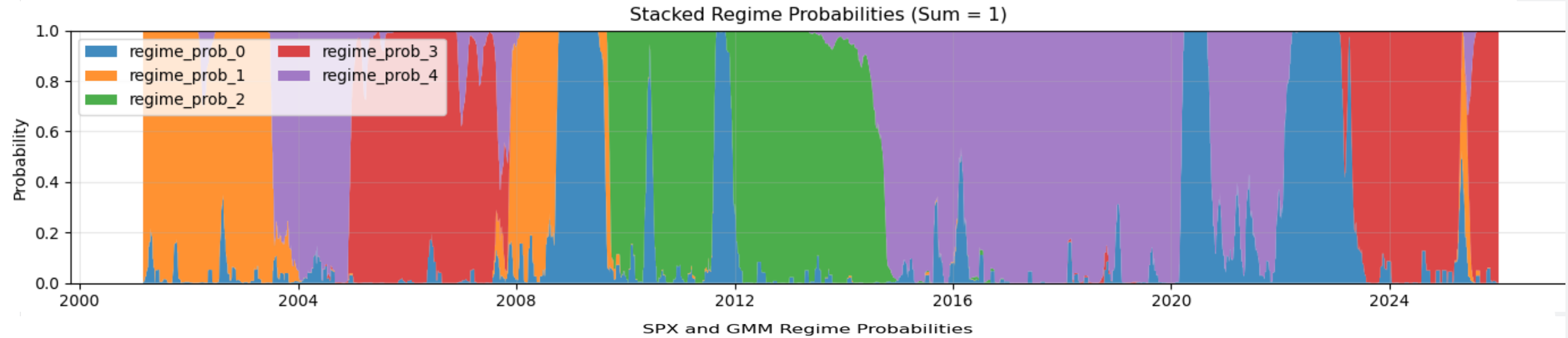


Silhouette Analysis for K Regimes



Metric	k=3	k=4	k=5
Avg. Silhouette	0.0945	0.0665	0.1174
Crisis Definition	Poorly defined / Mixed	Broad / Overlapping	Most Distinct
Model Utility	Too General	High Overlap	Optimal for Risk Signaling

Robust GMM for Regime Identification



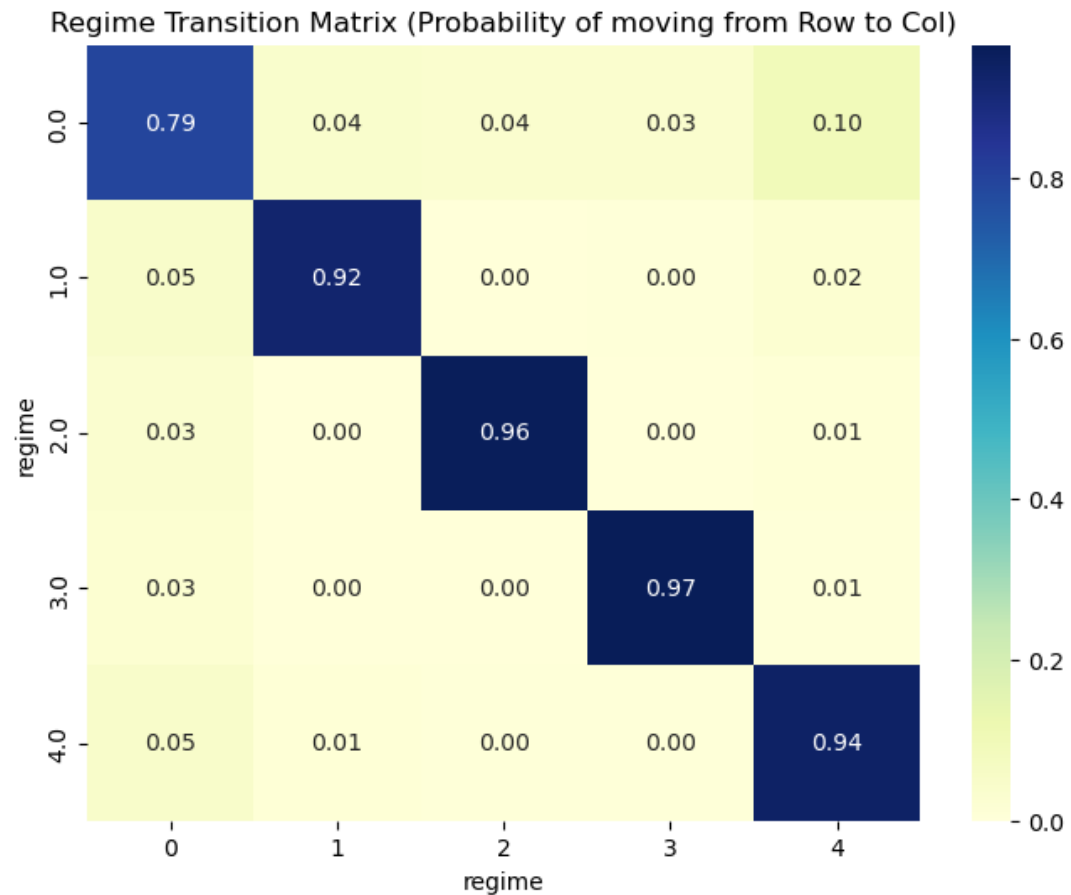
Economic Characterization of Identified Market Regimes

	regime	0	1	2	3	4
equity_market_dynamics_features_SPX_return		-0.0015	-0.0011	0.0011	0.0007	0.0010
Volatility_Risk_Sentiment_features_VIX		31.5064	25.3835	17.5770	14.5928	16.1602
credit_features_HY_IG_Spread		5.2940	5.7435	3.7462	2.2665	2.7926
Macro_Inflation_Environment_features_CPI_change_3m_lag1		0.6264	0.4964	0.3911	0.6410	0.4582
Interest_Rate_Environment_features_Slope_10_2		0.0096	0.0165	0.0215	0.0005	0.0108
CrossAsset_RiskSignals_features_Equity_bond_corr		-0.2834	-0.4624	-0.4586	0.0149	-0.2961

Regime	Name	SPX Return	VIX	Credit Spread	Yield Curve	Eq-Bond Corr
0	Crisis / Panic	↓ Negative	Very High (~31)	Wide	↘ Flat/Low	Negative
1	Recession / Credit Stress	↓ Negative	High (~25)	Highest	↗ Steepening	Very Negative
2	Expansion / Risk-On	↑ Strong Positive	Moderate (~17)	Tight	Steep	Negative
3	Late Cycle / Policy Tightening	↑ Positive	Low (~14)	Tightest	Flat	Positive
4	Goldilocks / Stable Growth	↑ Positive	Low (~16)	Tight	Normal	Negative



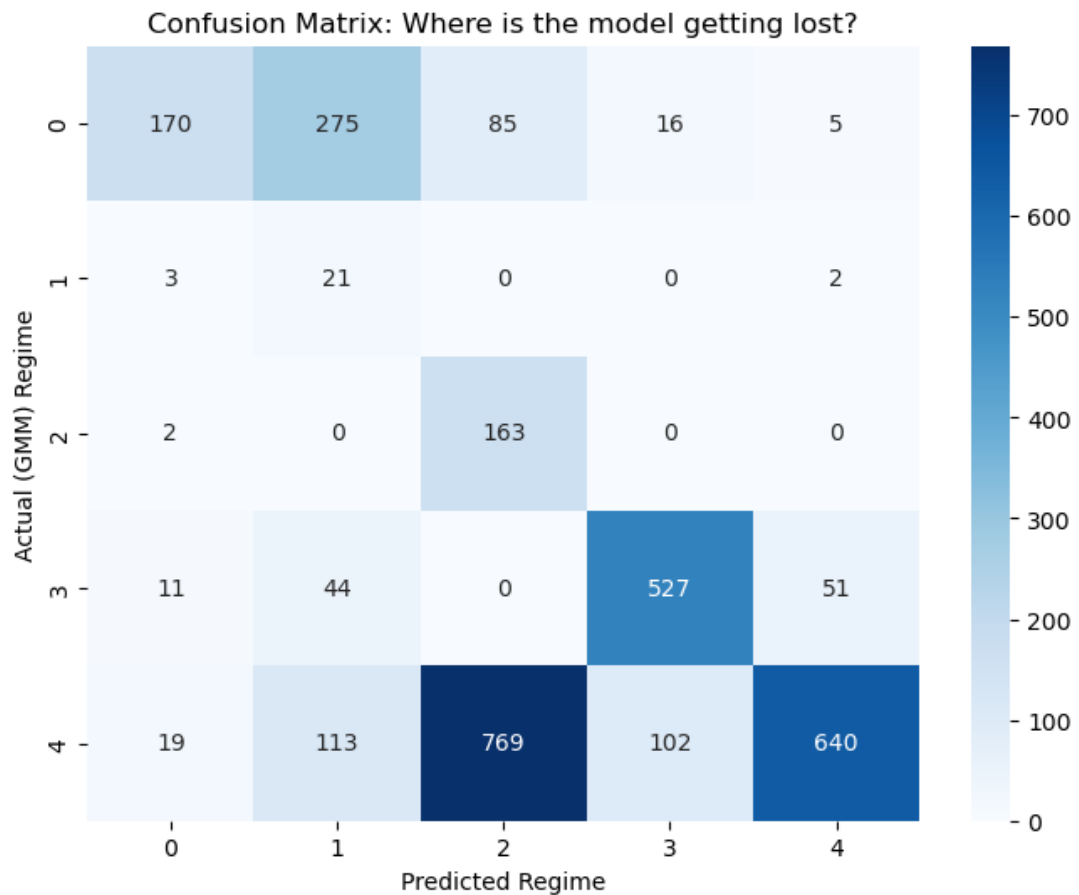
Regime Persistence and Transition Dynamics



- High Model Persistence
- Structural Longevity
- Crisis Velocity



Model Performance Diagnostic

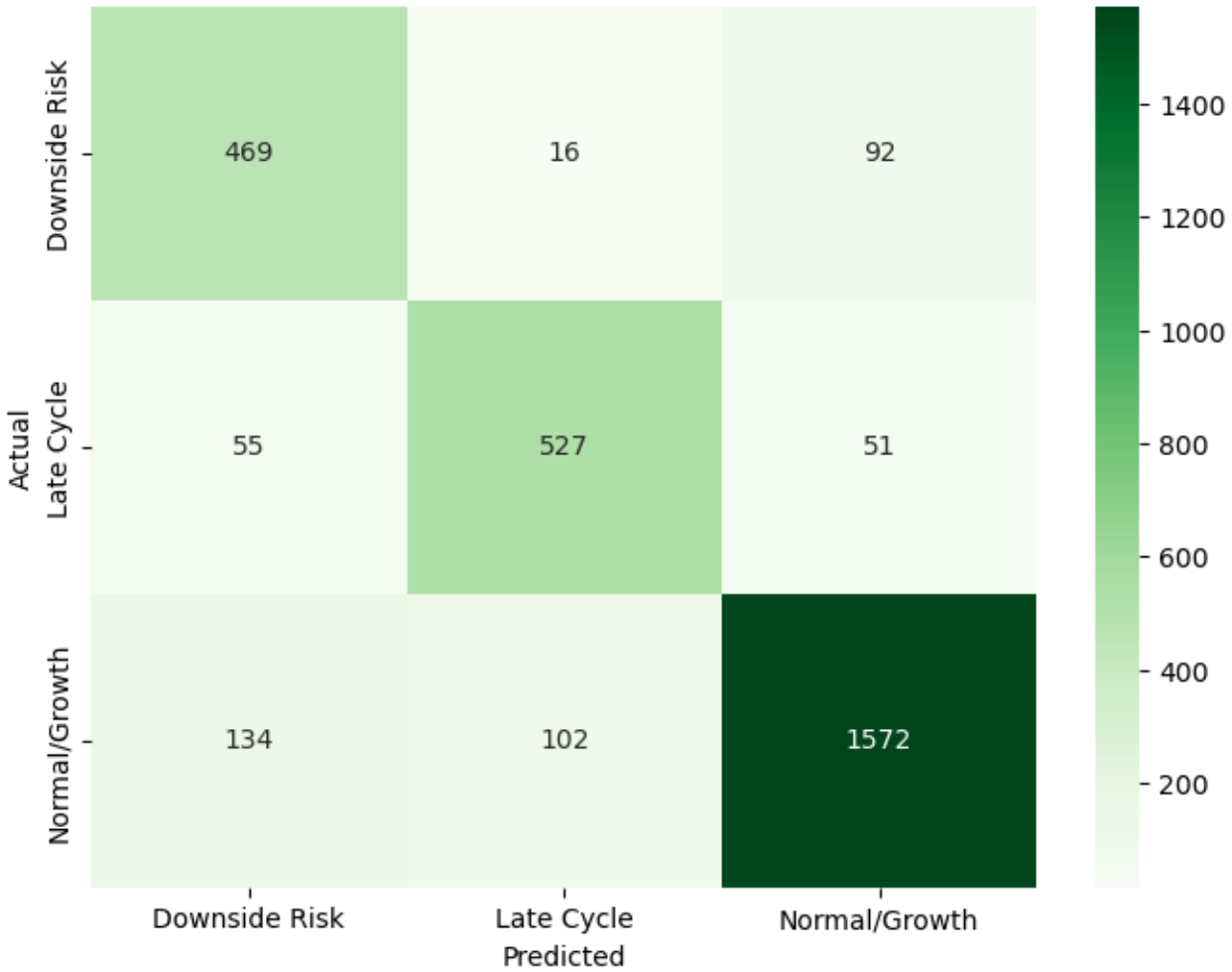


Regime	precision	recall	f1-score	support
0	0.83	0.31	0.45	551
1	0.05	0.81	0.09	26
2	0.16	0.99	0.28	165
3	0.82	0.83	0.82	633
4	0.92	0.39	0.55	1643

accuracy			0.5	3018
macro avg	0.55	0.67	0.44	3018
Weighted avg	0.83	0.5	0.57	3018

Merged Model Performance Diagnostic

New Confusion Matrix (Merged Regimes)



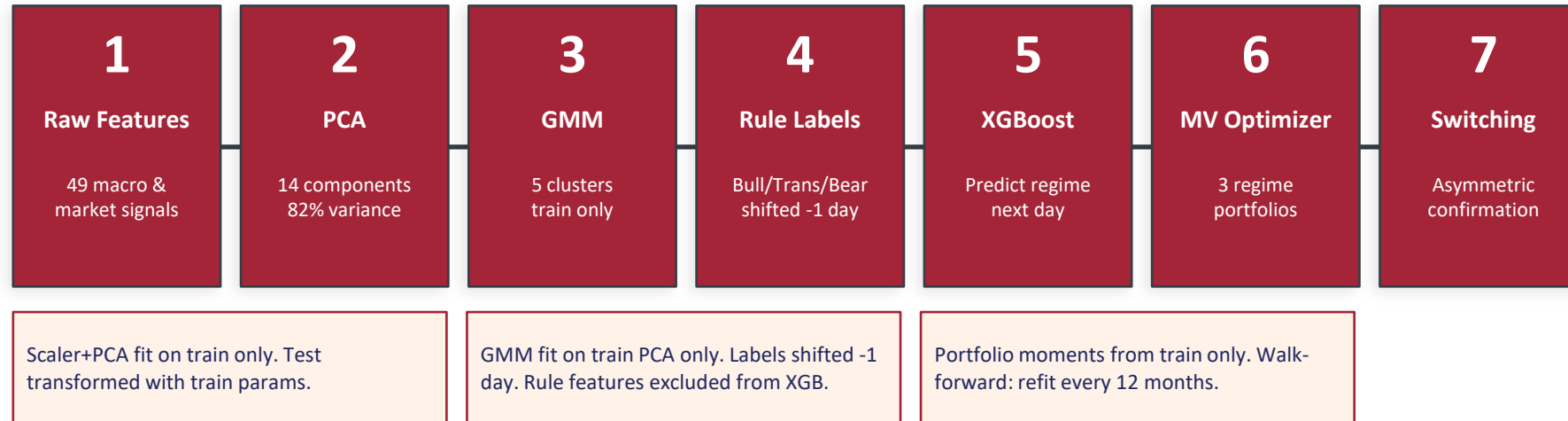
	precision	recall	f1-score	support
Downside Risk	0.71	0.81	0.76	577
Late Cycle	0.82	0.83	0.82	633
Normal/Growth	0.92	0.87	0.89	1808
accuracy			0.85	3018
macro avg	0.82	0.84	0.83	3018
weighted avg	0.86	0.85	0.85	3018

Merged Model Performance Gains

Metric	5-Class GMM	Merged 3-Class	Change
Weighted Accuracy	50%	85%	+35%
Signal Quality	High Noise	Actionable	Improved Reliability
Cost Efficiency	High Turnover	Low Turnover	Lower Tx Costs



Walk-Forward Pipeline Overview



Walk-Forward Expanding Window — Refit every 12 months

7 folds (2019–2025) · Each fold: train GMM+XGB on all data up to fold start · Predict next 12 months · No future data ever touches .fit()



Regime Labels — Rule-Based with Circularity Fix

BULL	Rule: VIX (5d smooth) < 18.0 Why: VIX < 18 = low fear, trending market. 18 is the historical boundary between calm expansion and elevated uncertainty.
TRANSITION	Rule: VIX (5d smooth) ≥ 18.0 AND not Bear Why: 18–25 VIX = elevated but not crisis. Warning zone — momentum mixed, credit spreads widening. Reduces conviction.
BEAR	Rule: VIX (5d smooth) ≥ 25.0 OR SPX drawdown < -12.5% Why: VIX ≥ 25 = genuine fear. Drawdown < -10% catches slow grinding bears even before VIX spikes.



XGBoost Classifier

Parameter	Explanation
n_estimators = 300	Stable out-of-sample; many trees ensure consensus across signals
max_depth = 4	Captures macro regime patterns; shallow trees reduce overfitting
learning_rate = 0.05	Gradual learning; prevents overfitting by limiting each tree's impact
subsample = 0.8	Uses 80% of data per tree; improves robustness and reduces outlier impact
class_weight = inverse freq	Upweights rare Bear regimes (~20%); improves crisis detection



Regime Switching & Execution Controls

PARAMETER	VALUE	WHY IT MATTERS
Switch Threshold	0.75	Only act on high-confidence signals → reduces whipsaw
Min Probability Gap	0.15	Avoid switching when regimes are ambiguous
Enter Confirmation	7 days	Prevents reacting to short-term noise
Exit Confirmation	1 day	Faster exit from risk (asymmetric design)
Min Hold (Bull / Trans / Bear)	21 / 15 / 10 days	Enforces persistence; reduces turnover
Transaction Cost	5 bps	Ensures realistic, net-of-cost performance



Constrained MV Optimizer

Global (Non-Regime Aware) Constraints

$$\text{maximize } \mu_s^T w - \gamma_s(w^T \Sigma_s w) - \lambda \|w\|_2^2$$

Constraint	Value	Explanation
Max Asset Weight	40%	Prevents concentration risk
Max Active Tilt	50%	Limits deviation from benchmark
Group Tilt Limit	50%	Avoids extreme asset class shifts
L2 Regularization	0.00005	Stabilizes weights; avoids extreme positions
Transaction Cost	5 bps	Ensures realistic portfolio performance



Constrained MV Optimizer

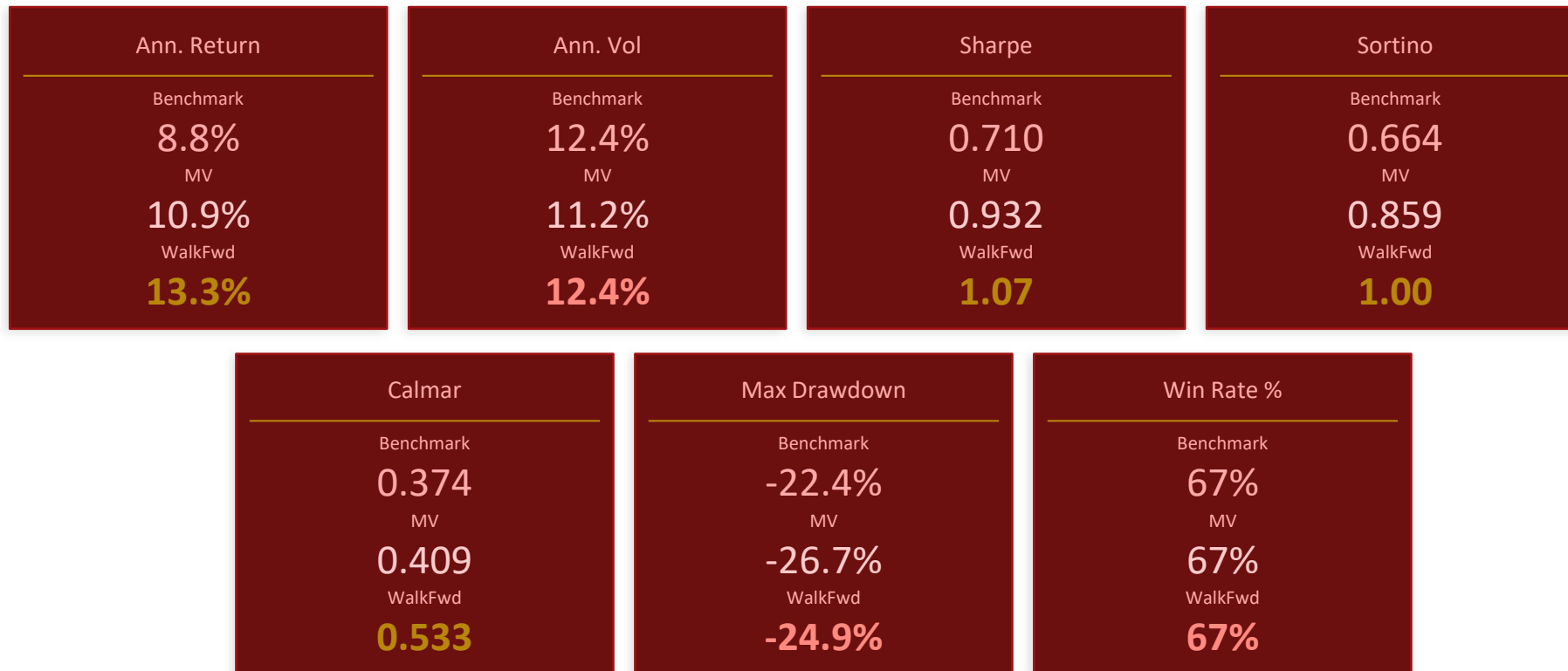
Regime-Specific Constraints

Parameter	Bull	Transition	Bear	Explanation
Risk Aversion (γ)	1.0	2.0	4.0	Adjusts risk-taking across regimes
Max Active Tilt	35%	25%	20%	More conservative positioning in Bear
Group Tilt Limit	40%	30%	25%	Limits extreme allocation shifts
Equity Range	60% – 80%	45% – 65%	None	Forces risk-taking in Bull; flexibility in Bear
Bond Allocation	10% – 40%	20% – 45%	None	Balances diversification vs flexibility
Alt (Gold) Cap	$\leq 20\%$	$\leq 25\%$	$\leq 20\%$	Controls alternative exposure
Duration Control (Bear)	—	—	Cap TLT/IEF	Reduces bond losses in rising yields



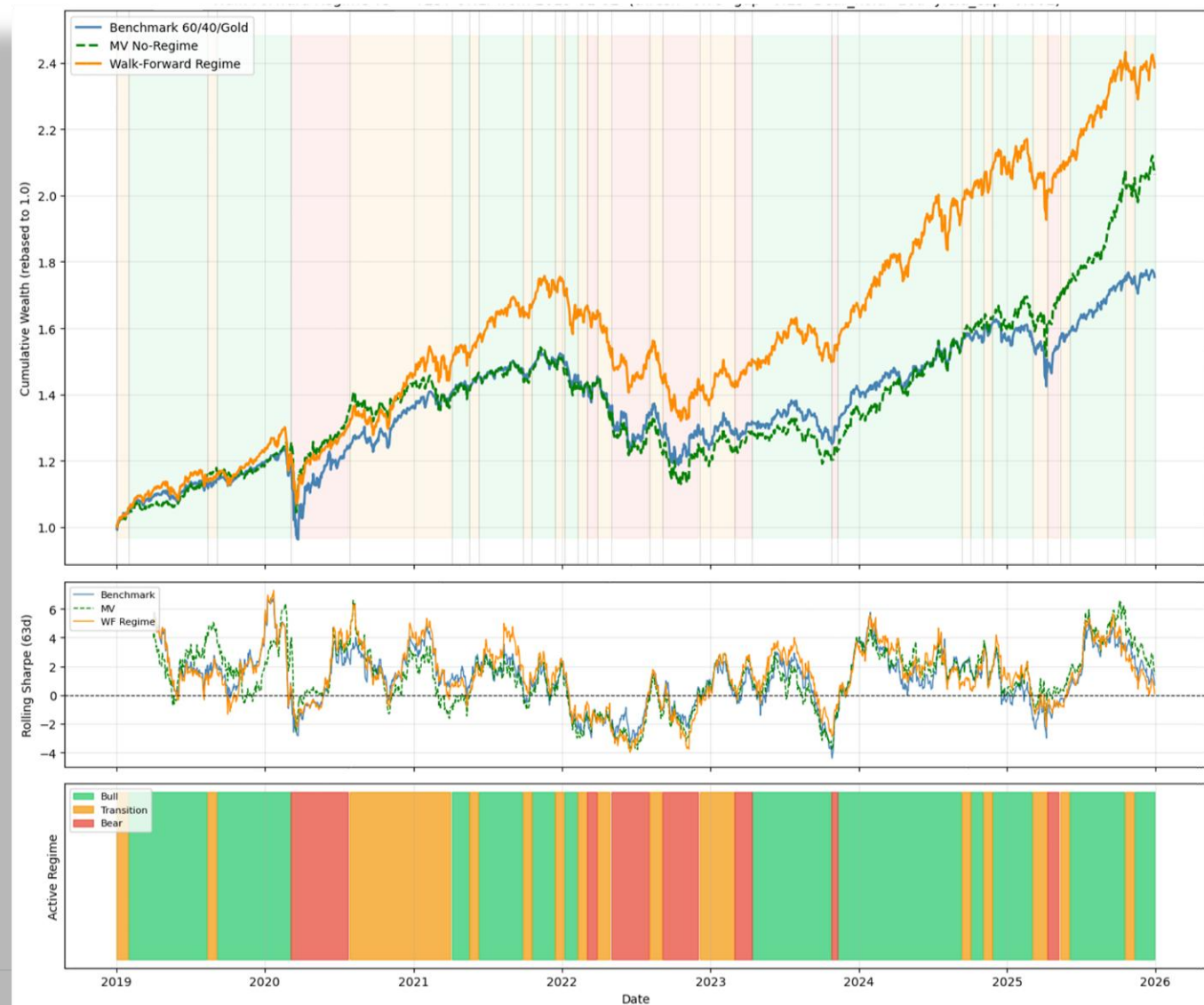
Full Performance Summary

Test Period: 2019–2025 · 7 years · 7 walk-forward folds · TC = 5bps per trade



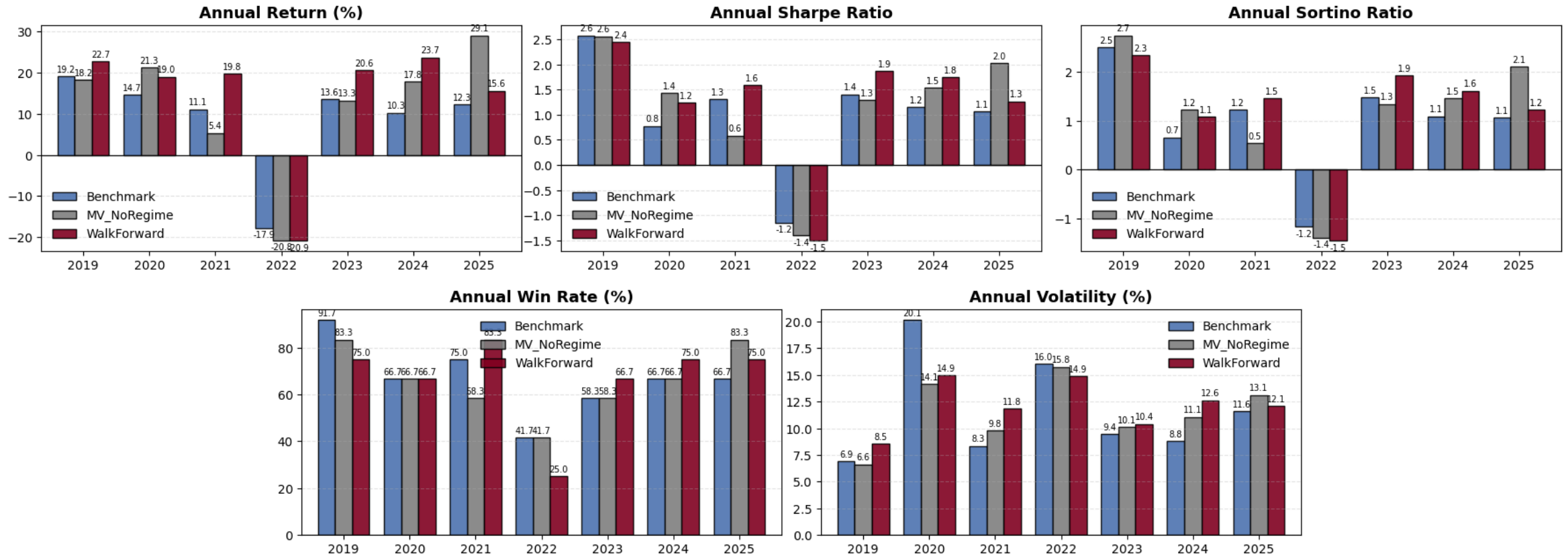
Performance Analysis

- Significant Outperformance
- Effective Downside Protection
- Dynamic Adaptation
- Higher Risk-Efficiency
- Model Validation



Annual Return Comparison (2019–2025)

Annual Performance Metrics



Regime Alpha Table - What Each Regime Contributes?

Regime	Days	Regime Ret	Bmark Ret	Alpha	Regime Sharpe	Bmark Sharpe	Sharpe Diff
Bull	988.0	15.66%	8.20%	↑ +7.46%	1.324	1.000	↑ +0.324
Transition	467.0	8.51%	4.48%	↑ +4.03%	0.705	0.420	↑ +0.285
Bear	305.0	12.96%	15.24%	↓ -2.28%	0.912	0.777	↑ +0.135

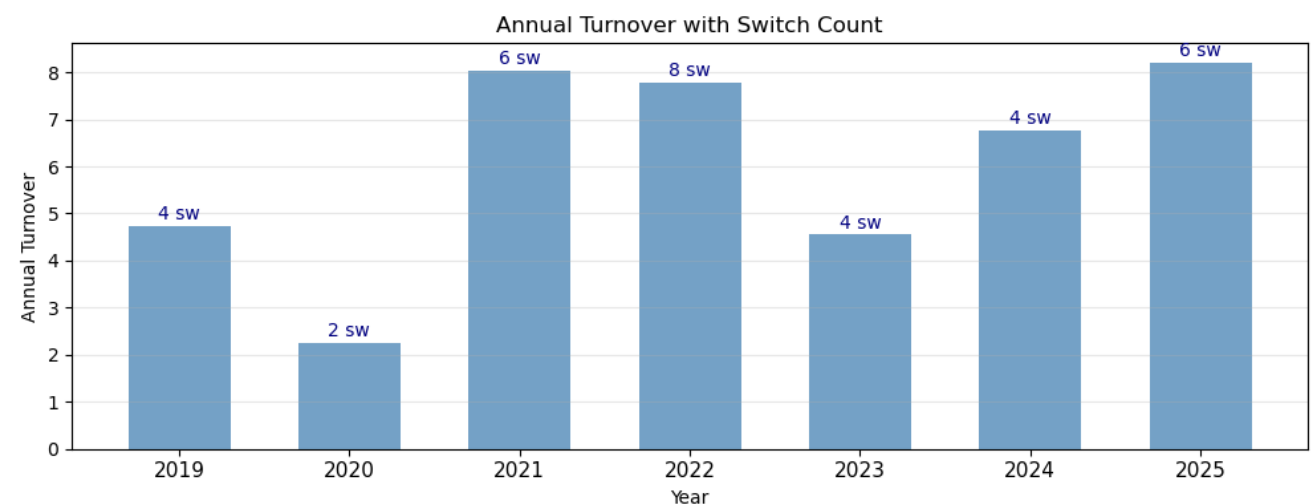
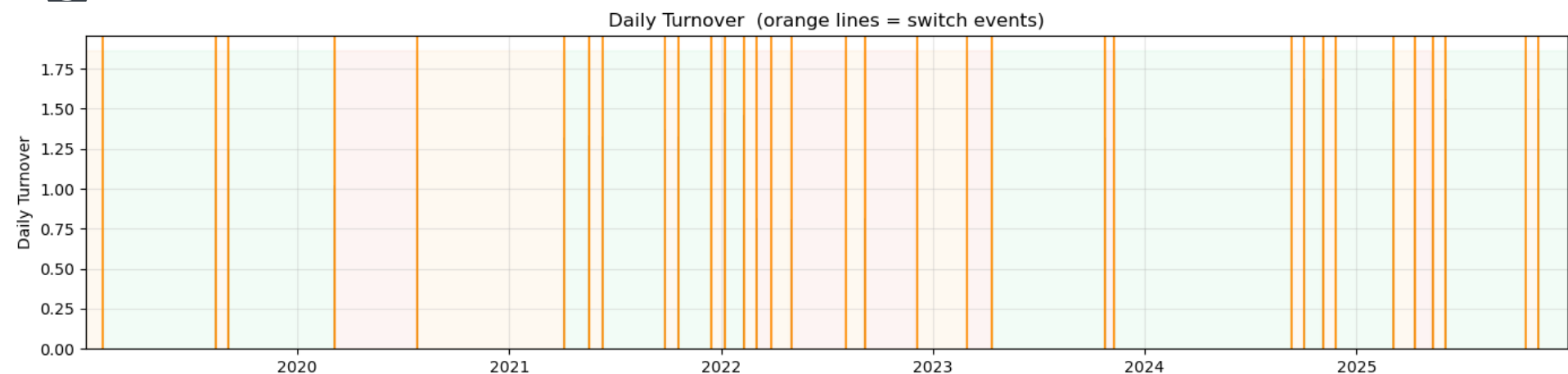
BULL: +7.46% alpha · Sharpe 1.324 — equity_min=60% + $\gamma=1.0$ drives aggressive tilt. 7-day confirm ensures clean trend entry.

TRANSITION: +4.03% alpha — balanced equity (45–65%) keeps portfolio productive during uncertainty.

BEAR: -2.28% alpha but better risk adjusted return



ANNUAL TURNOVER — Walk-Forward Regime



Year	Switches	Total Turnover
2019	4	4.725
2020	2	2.261
2021	6	8.027
2022	8	7.776
2023	4	4.547
2024	4	6.756
2025	6	8.204
TOTAL	34	42.296

Transaction Cost

Turnover per Switch:

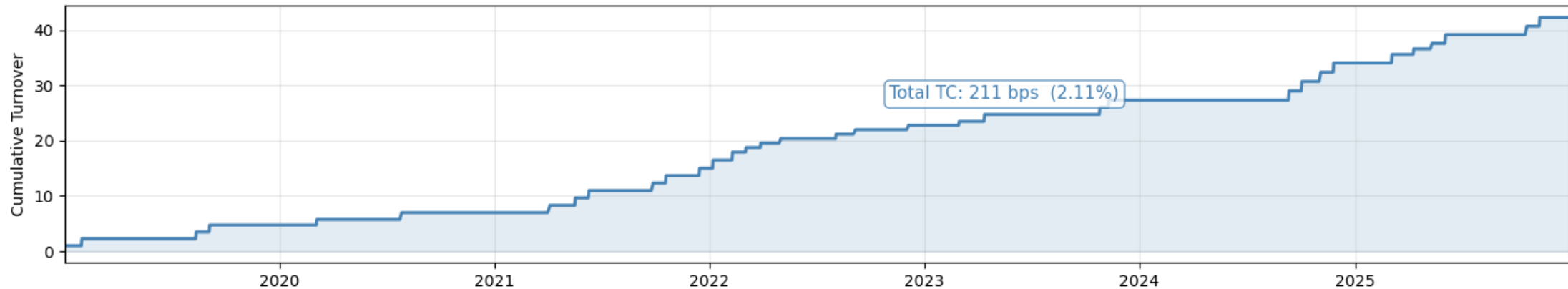
$$\tau_t = \sum_{i=1}^N |w_i^{new} - w_i^{old}|$$

Transaction Cost per Switch:

$$TC_t = \frac{c}{10,000} \times \tau_t \quad \text{where } c = 5 \text{ bps}$$

$$r_t^{net} = \mathbf{w}_t^\top \mathbf{r}_t - \frac{5}{10,000} \sum_{i=1}^N |w_{i,t}^{new} - w_{i,t}^{old}|$$

Cumulative Turnover



Limitations & Future Roadmap

- **2022 rate-hike bear missing from training**
 - Train window (2001–2018) contains no sustained period where stocks AND bonds fell together. Model entered Bear correctly but TLT/IEF losses hurt. Yield-cap (v5) partially addresses this.
- **Short 7-year test window**
 - 2019–2025 is predominantly bullish. A full cycle including 2000–2002 or 2008–2009 would provide stronger out-of-sample evidence. Bear regime value is likely underestimated.
- **Bear benchmark outperforms**
 - Benchmark earns +13.8% during Bear periods — mainly the 2020 snap-back rally (+60% April–December). Strategy exits Bear correctly but the defensive portfolio misses the recovery.
- **Parameter calibration on test period**
 - SWITCH_THRESHOLD=0.75 and MIN_PROB_GAP=0.15 were refined after seeing test data. True blind out-of-sample would require fixing these before observing any test returns.
- **Bear moment blending is heuristic**
 - 60/40 blend of Bear-regime vs full-history covariance is undocumented. Small Bear samples make estimates noisy; a formal shrinkage estimator would be more principled.
- **4th regime: stagflation**
 - Explicit StagBear regime (high VIX + rising yields). Achievable once 2022 enters the training window — the 2023 refit already sees early rising rates in its history.
- **DCC-GARCH covariance**
 - Replace static regime covariance with dynamic conditional covariance. Captures volatility clustering within regimes more accurately, especially around crisis onset.
- **TIPS & commodities**
 - Add inflation-hedge assets for Bear regime. TIPS and commodity ETFs (DJP) provide genuine diversification when both SPY and TLT fail simultaneously.
- **Compare the ML model with HHM, HSHM, Jump Model**
 - Replace GMM with Hidden Markov Model or LSTM to improve temporal consistency. Target: raise Crisis/Panic recall from 31% without sacrificing Goldilocks precision





THANK YOU

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